



AD23431 - STATISTICAL ANALYSIS AND COMPUTING

EXP 6: Loading of Files and Data preprocessing in R

A healthcare organization maintains patient health records for medical analytics and reporting. The data is stored across two different sources:

- **Outpatient records** → CSV file
- **Inpatient records** → Relational database (SQLite)

Each dataset contains the following attributes:

- **PatientID** – Unique identifier for each patient
- **Age** – Numerical attribute representing patient age
- **BloodPressure** – Numerical attribute representing systolic blood pressure
- **Diagnosis** – Categorical attribute indicating disease type (Diabetes, Hypertension, Cardiac, Normal)
- **HospitalBranch** – Categorical attribute representing hospital location

Due to manual data entry errors and system synchronization issues, the datasets contain:

- Missing values
- Duplicate records
- Inconsistent categorical formats (e.g., "diabetes", "Diabetes", "DIABETES")
- Data type inconsistencies

The organization needs to load both datasets into R, integrate them, explore the structure, separate numerical and categorical attributes, clean inconsistencies, and prepare a consistent dataset for medical analysis.

Objective:

To load patient health data from a CSV file and a relational database into R, integrate the datasets, explore the data, separate numerical and categorical attributes, handle missing values and duplicates, standardize categorical variables, convert them into factors, and prepare a clean dataset for healthcare analytics.



Procedure:

Step 1: Load Required Packages

Step 2: Create and Store CSV Dataset (Outpatient Records)

- Create sample outpatient dataset
- Save as CSV file

Step 3: Create and Store SQLite Dataset (Inpatient Records)

- Create SQLite database
- Insert inpatient records

Step 4: Load Both Datasets into R

- Import CSV file
- Connect to SQLite database

Step 5: Integrate Datasets

- Combine outpatient and inpatient records. Resolve schema conflicts
- Print integrated dataset `patient_data`

Step 6: Data Exploration

- Display structure using `str()`
- Generate summary statistics using `summary()`
- Count missing values column-wise
- Identify duplicate records
- Display frequency distribution of Diagnosis and HospitalBranch using `table()`

Step 7: Separate Numerical and Categorical Data

- Extract numerical attributes (Age, BloodPressure)
- Extract categorical attributes (Diagnosis, HospitalBranch)
- Print both subsets

Step 8: Data Cleaning

- Handle missing numerical values using mean/median
- Handle missing categorical values using mode
- Standardize inconsistent categorical values
- Remove duplicate records
- Convert categorical variables into factors
- Print final cleaned dataset



```
install.packages("DBI")
install.packages("RSQLite")
install.packages("dplyr")

library(DBI)
library(RSQLite)
library(dplyr)

outpatient <- data.frame(
  PatientID=c(101,102,103,104,105,106,106),
  Age=c(25,40,NA,35,50,60,60),
  BloodPressure=c(120,140,130,NA,150,160,160),
  Diagnosis=c("Diabetes","hypertension","DIABETES","Normal","Cardiac",NA,
"Cardiac"),
  HospitalBranch=c("Chennai","chennai","Madurai","Coimbatore","Madurai","C
hennai","Chennai")
)

write.csv(outpatient,"outpatient_records.csv",row.names=FALSE)

con <- dbConnect(SQLite(),"hospital.db")

inpatient <- data.frame(
  PatientID=c(201,202,203,204,205,101),
  Age=c(45,55,65,NA,70,25),
  BloodPressure=c(145,155,NA,135,165,120),
  Diagnosis=c("cardiac","Diabetes","Hypertension","Normal","DIABETES","Dia
betes"),
  HospitalBranch=c("Chennai","Madurai","Madurai","Coimbatore","Chennai","C
hennai"))

dbWriteTable(con,"inpatient_records",inpatient,overwrite=TRUE)

outpatient_data <- read.csv("outpatient_records.csv",stringsAsFactors=FALSE)
inpatient_data <- dbReadTable(con,"inpatient_records")

patient_data <- bind_rows(outpatient_data,inpatient_data)
```



```
str(patient_data)
```

Output:

```
'data.frame': 13 obs. of 5 variables:
```

```
$ PatientID : num 101 102 103 104 105 106 106 201 202 203 ...
```

```
$ Age : num 25 40 NA 35 50 60 60 45 55 65 ...
```

```
$ BloodPressure : num 120 140 130 NA 150 160 160 145 155 NA ...
```

```
$ Diagnosis : chr "Diabetes" "hypertension" "DIABETES" "Normal"
```

```
...
```

```
$ HospitalBranch: chr "Chennai" "chennai" "Madurai" "Coimbatore" ...
```

```
summary(patient_data)
```

Output:

```
PatientID Age BloodPressure Diagnosis
```

```
Min. :101.0 Min. :25.00 Min. :120.0 Length:13
```

```
1st Qu.:103.0 1st Qu.:37.50 1st Qu.:132.5 Class :character
```

```
Median :106.0 Median :50.00 Median :145.0 Mode :character
```

```
Mean :141.8 Mean :48.18 Mean :143.6
```

```
3rd Qu.:202.0 3rd Qu.:60.00 3rd Qu.:157.5
```

```
Max. :205.0 Max. :70.00 Max. :165.0
```

```
NA's :2 NA's :2
```

```
HospitalBranch
```

```
Length:13
```

```
Class :character
```

```
Mode :character
```

```
colSums(is.na(patient_data))
```

Output:

```
PatientID Age BloodPressure Diagnosis HospitalBranch
0 2 2 1 0
```

```
patient_data[duplicated(patient_data),]
```

Output:

```
PatientID Age BloodPressure Diagnosis HospitalBranch
13 101 25 120 Diabetes Chennai
```



```
table(patient_data$Diagnosis)
```

Output:

cardiac	Cardiac	Diabetes	DIABETES	hypertension	Hypertension
1	2	3	2	1	1
Normal					
2					

```
table(patient_data$HospitalBranch)
```

Output:

chennai	Chennai	Coimbatore	Madurai
1	6	2	4

```
numerical_data <- select(patient_data, Age, BloodPressure)
```

```
categorical_data <- select(patient_data, Diagnosis, HospitalBranch)
```

```
numerical_data
```

Output:

	Age	BloodPressure
1	25	120
2	40	140
3	NA	130
4	35	NA
5	50	150
6	60	160
7	60	160
8	45	145
9	55	155
10	65	NA
11	NA	135
12	70	165
13	25	120



categorical_data

Output:

	Diagnosis	HospitalBranch
1	Diabetes	Chennai
2	hypertension	chennai
3	DIABETES	Madurai
4	Normal	Coimbatore
5	Cardiac	Madurai
6	<NA>	Chennai
7	Cardiac	Chennai
8	cardiac	Chennai
9	Diabetes	Madurai
10	Hypertension	Madurai
11	Normal	Coimbatore
12	DIABETES	Chennai
13	Diabetes	Chennai

```
patient_data$Age[is.na(patient_data$Age)]
mean(patient_data$Age,na.rm=TRUE)
patient_data$BloodPressure[is.na(patient_data$BloodPressure)]
mean(patient_data$BloodPressure,na.rm=TRUE)
```

```
mode_fun
function(x){ux<-unique(x);ux[which.max(tabulate(match(x,ux)))]}
```

```
patient_data$Diagnosis[is.na(patient_data$Diagnosis)]
mode_fun(patient_data$Diagnosis)
```

```
patient_data$Diagnosis <- tools::toTitleCase(tolower(patient_data$Diagnosis))
patient_data$HospitalBranchtools::toTitleCase(tolower(patient_data$HospitalBranch))
```

```
patient_data <- distinct(patient_data)
```

```
patient_data$Diagnosis <- as.factor(patient_data$Diagnosis)
patient_data$HospitalBranch <- as.factor(patient_data$HospitalBranch)
```



patient_data

Output:

	PatientID	Age	BloodPressure	Diagnosis	HospitalBranch
1	101	25.00000	120.0000	Diabetes	Chennai
2	102	40.00000	140.0000	Hypertension	Chennai
3	103	48.18182	130.0000	Diabetes	Madurai
4	104	35.00000	143.6364	Normal	Coimbatore
5	105	50.00000	150.0000	Cardiac	Madurai
6	106	60.00000	160.0000	Diabetes	Chennai
7	106	60.00000	160.0000	Cardiac	Chennai
8	201	45.00000	145.0000	Cardiac	Chennai
9	202	55.00000	155.0000	Diabetes	Madurai
10	203	65.00000	143.6364	Hypertension	Madurai
11	204	48.18182	135.0000	Normal	Coimbatore
12	205	70.00000	165.0000	Diabetes	Chennai

dbDisconnect(con)